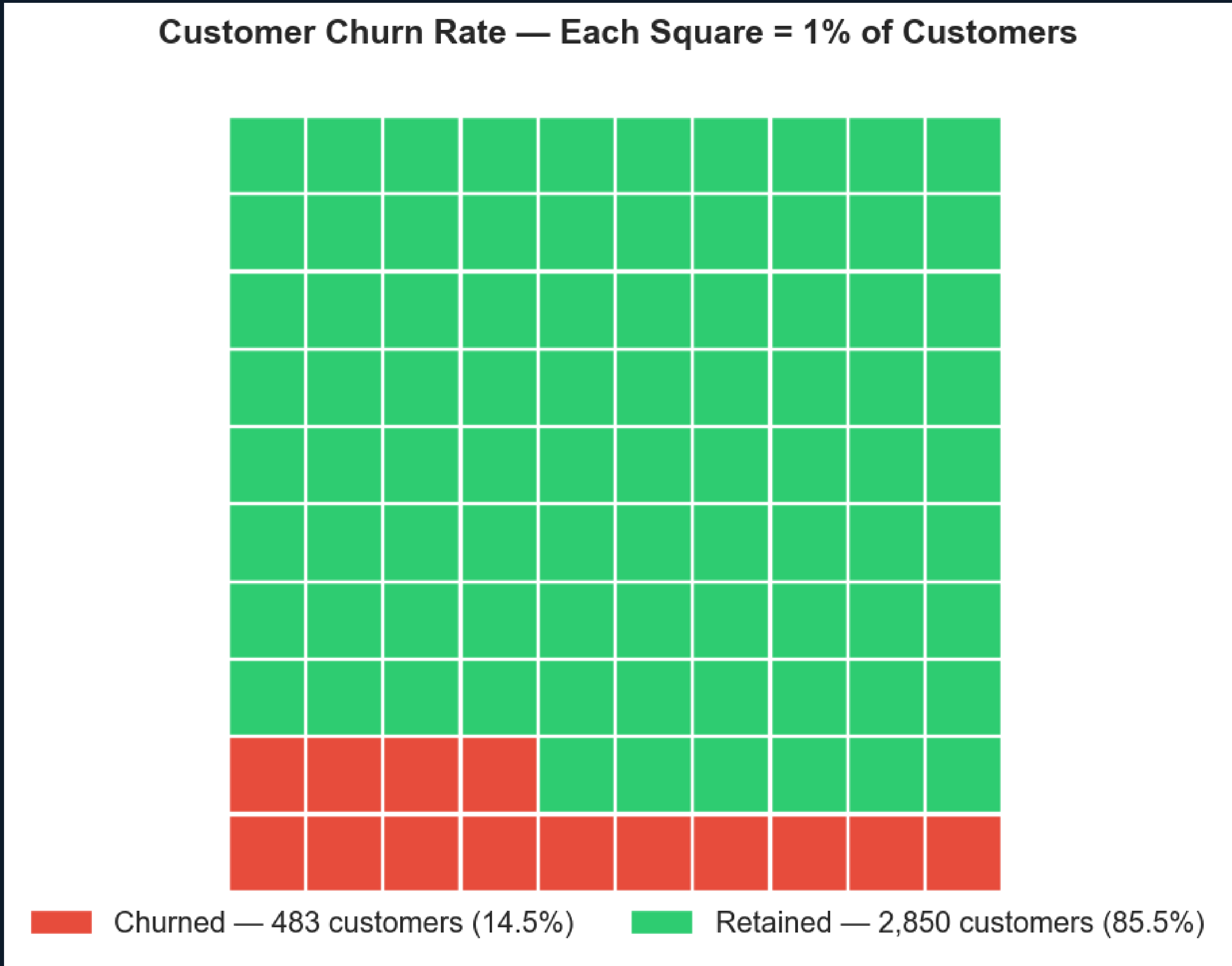
Telecom Customer Churn Prediction

All 18 Notebook Visuals

EDA · Modeling · Evaluation

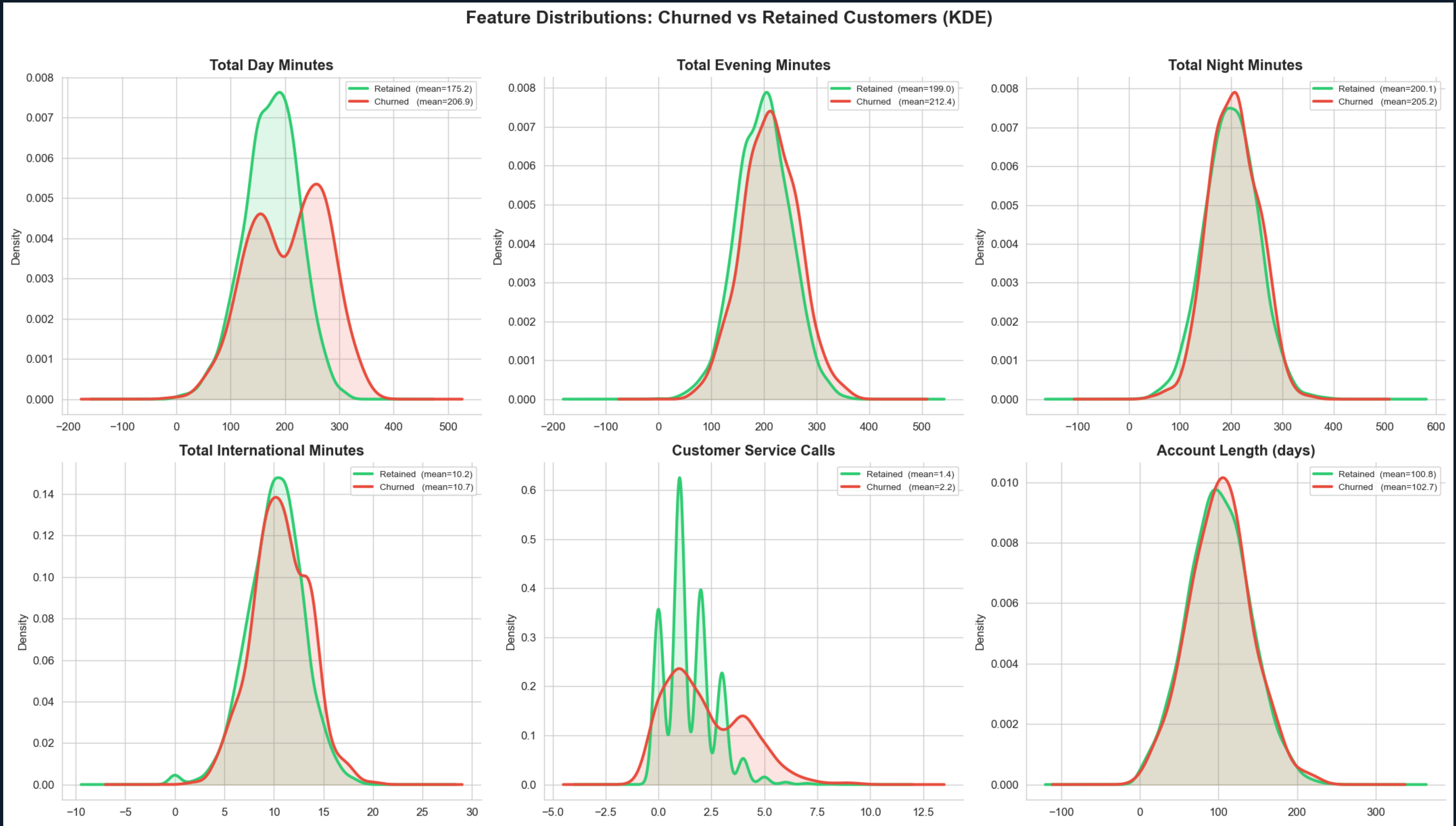
Phase 3 Machine Learning Project · CRISP-DM Framework

■■ EXPLORATORY DATA ANALYSIS ■■



Visual 1 — Waffle Chart: Churn Rate Each square = 1% of customers. 14 red squares = 14.49% annual churn rate.

EXPLORATORY DATA ANALYSIS

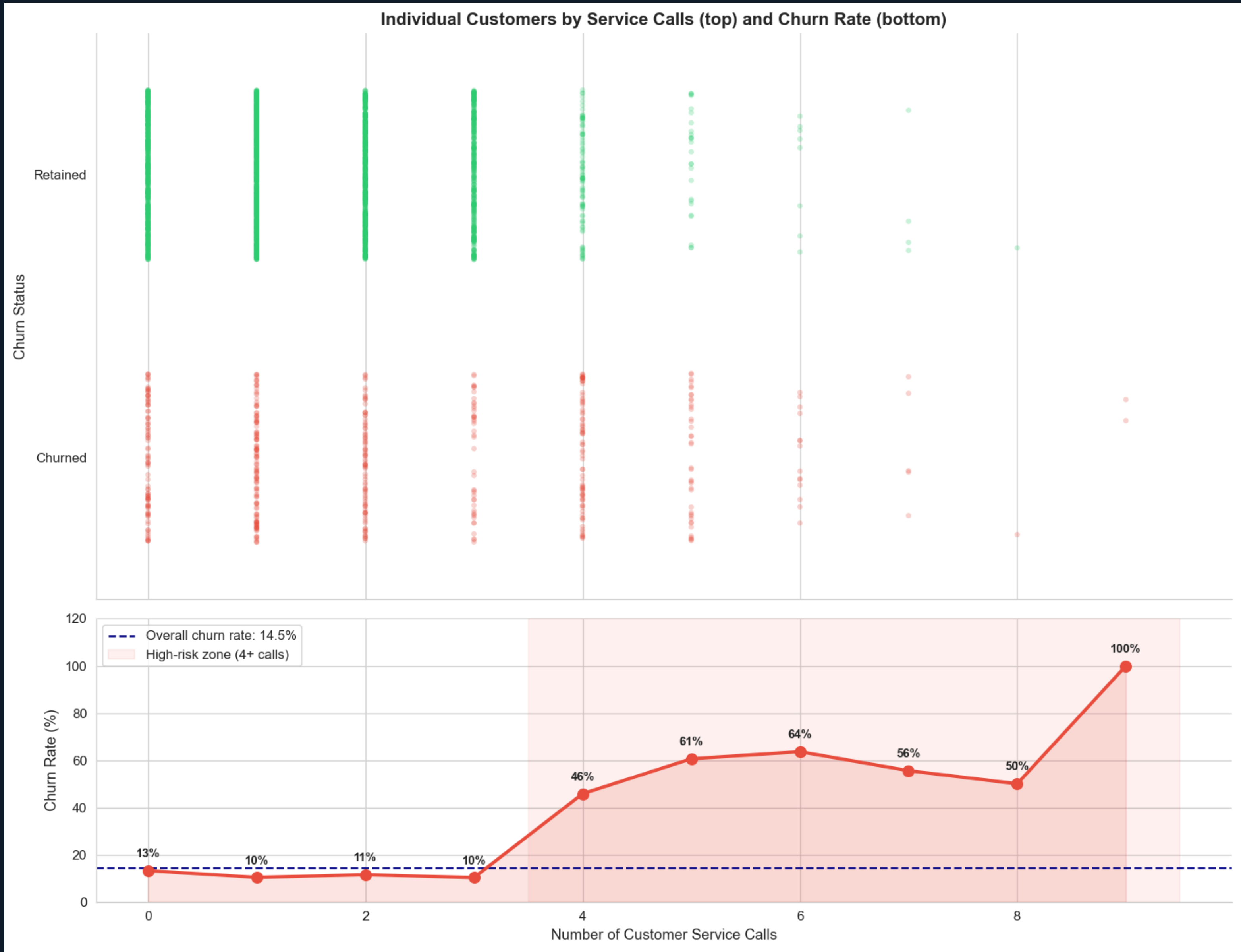


Visual 2 — KDE Plots: Usage Distributions by Churn Status Churned customers average 207 vs 175 day minutes for retained — higher usage drives price sensitivity.

EXPLORATORY DATA ANALYSIS

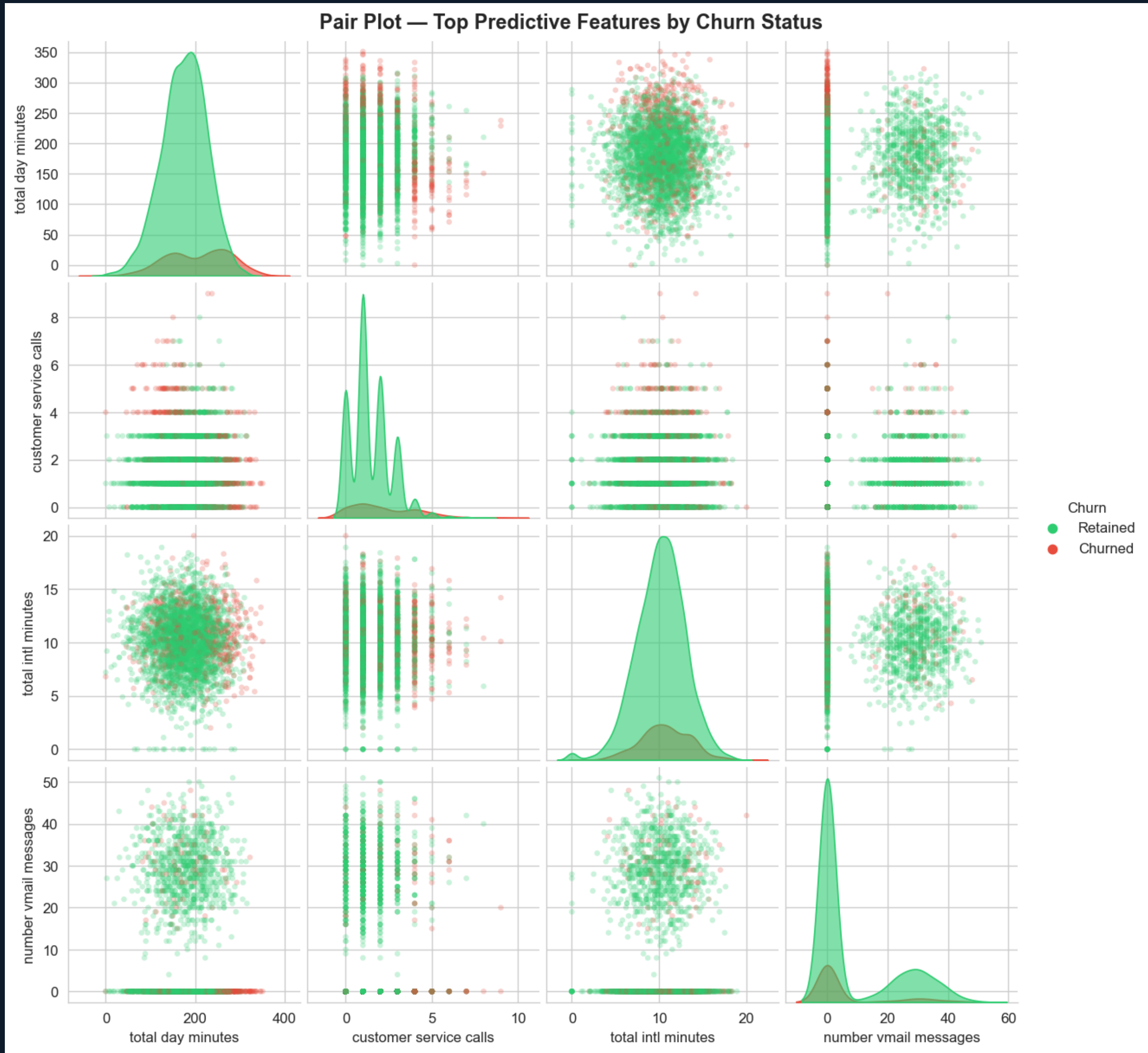
Visual 3 — Violin Plots: Usage Spread by Time-of-Day and Churn Combines box plot and KDE to show the full distribution shape for each usage period.

EXPLORATORY DATA ANALYSIS



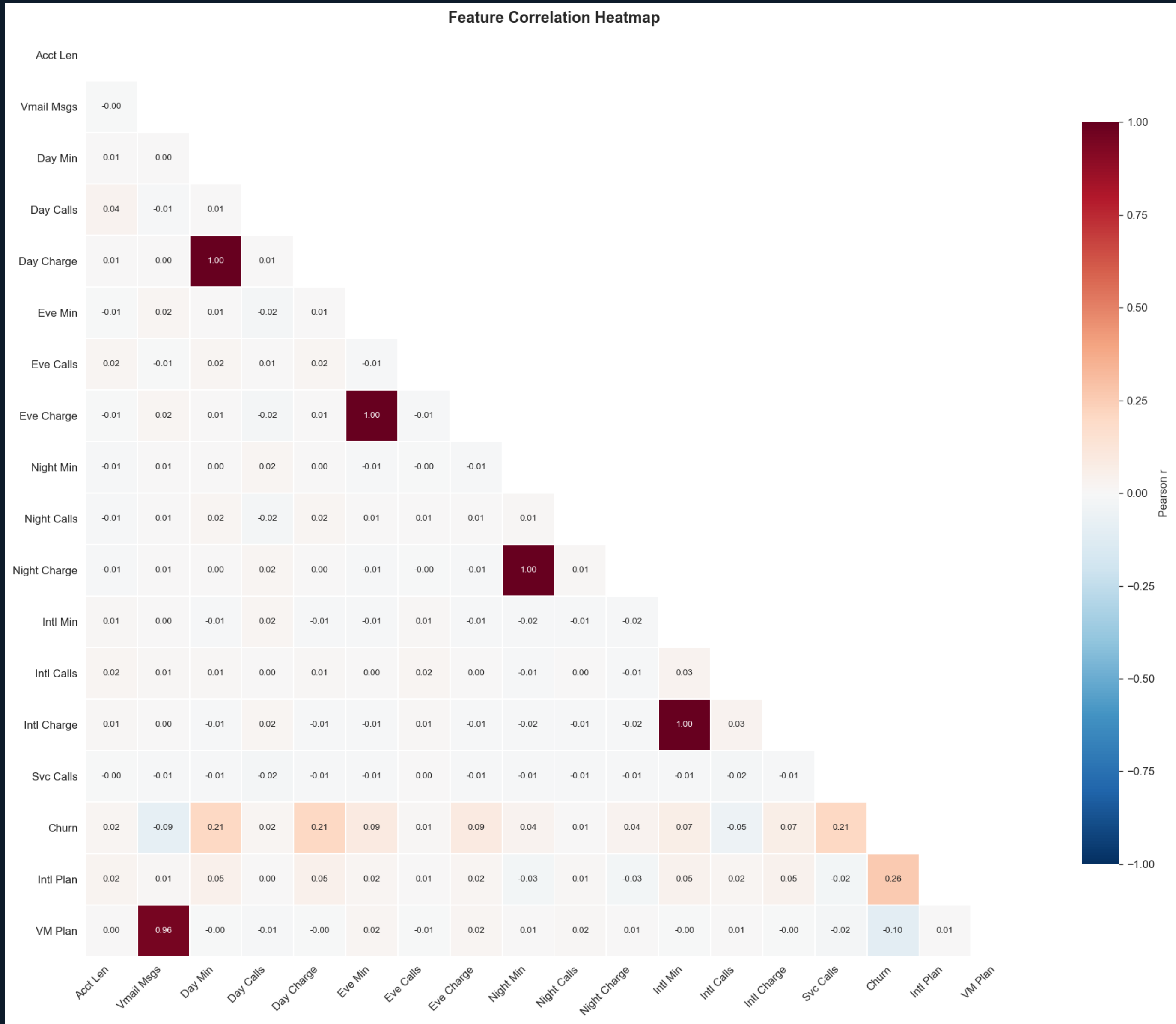
Visual 4 — Strip Plot + Churn Rate Overlay: Customer Service Calls Churn rate jumps from ~11% (0–3 calls) to 46%+ at 4+ calls — a clear non-linear threshold effect.

EXPLORATORY DATA ANALYSIS

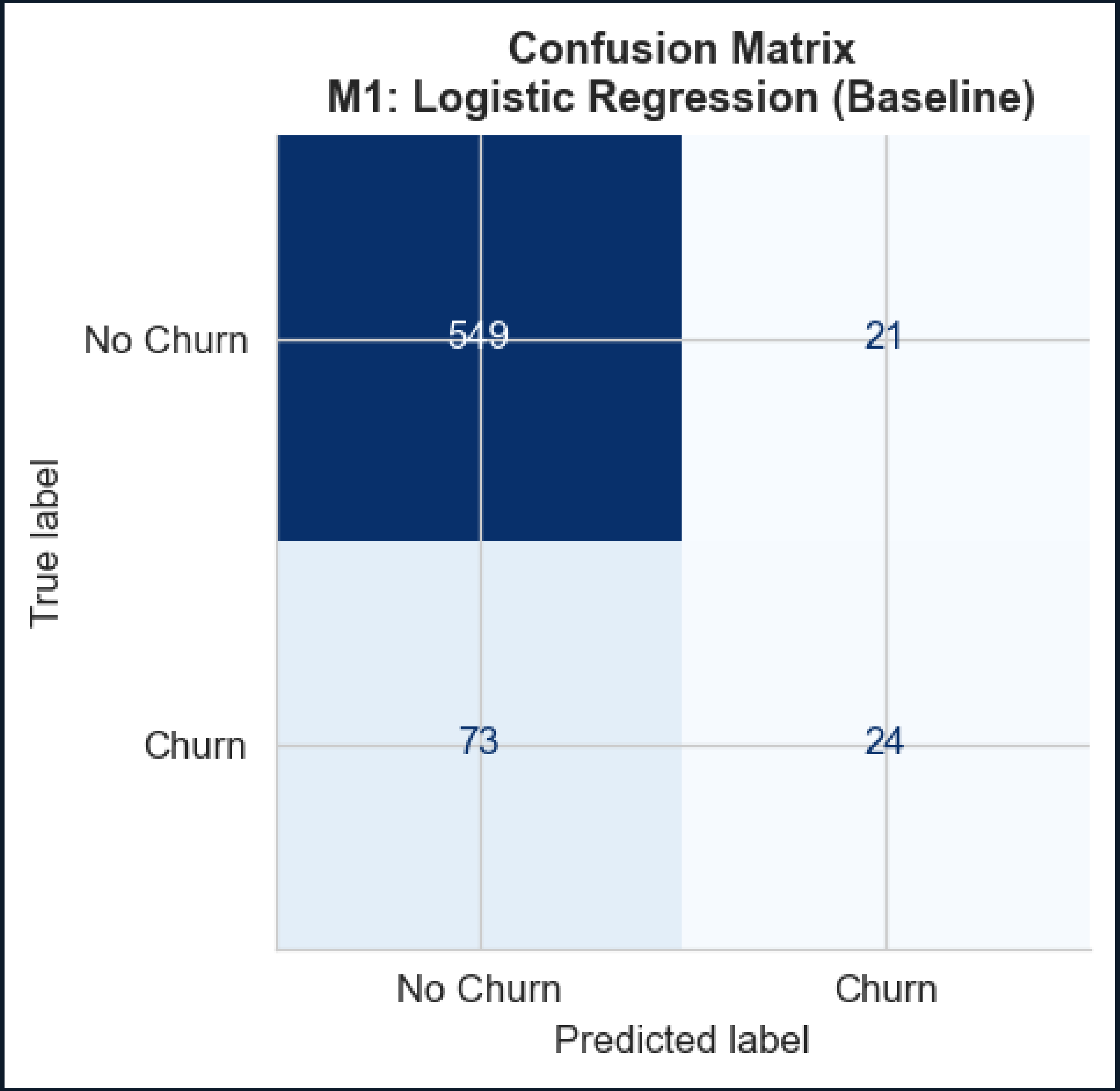


Visual 5 — Pair Plot: Top Predictive Features Pairwise relationships between the top 4 features coloured by churn status.

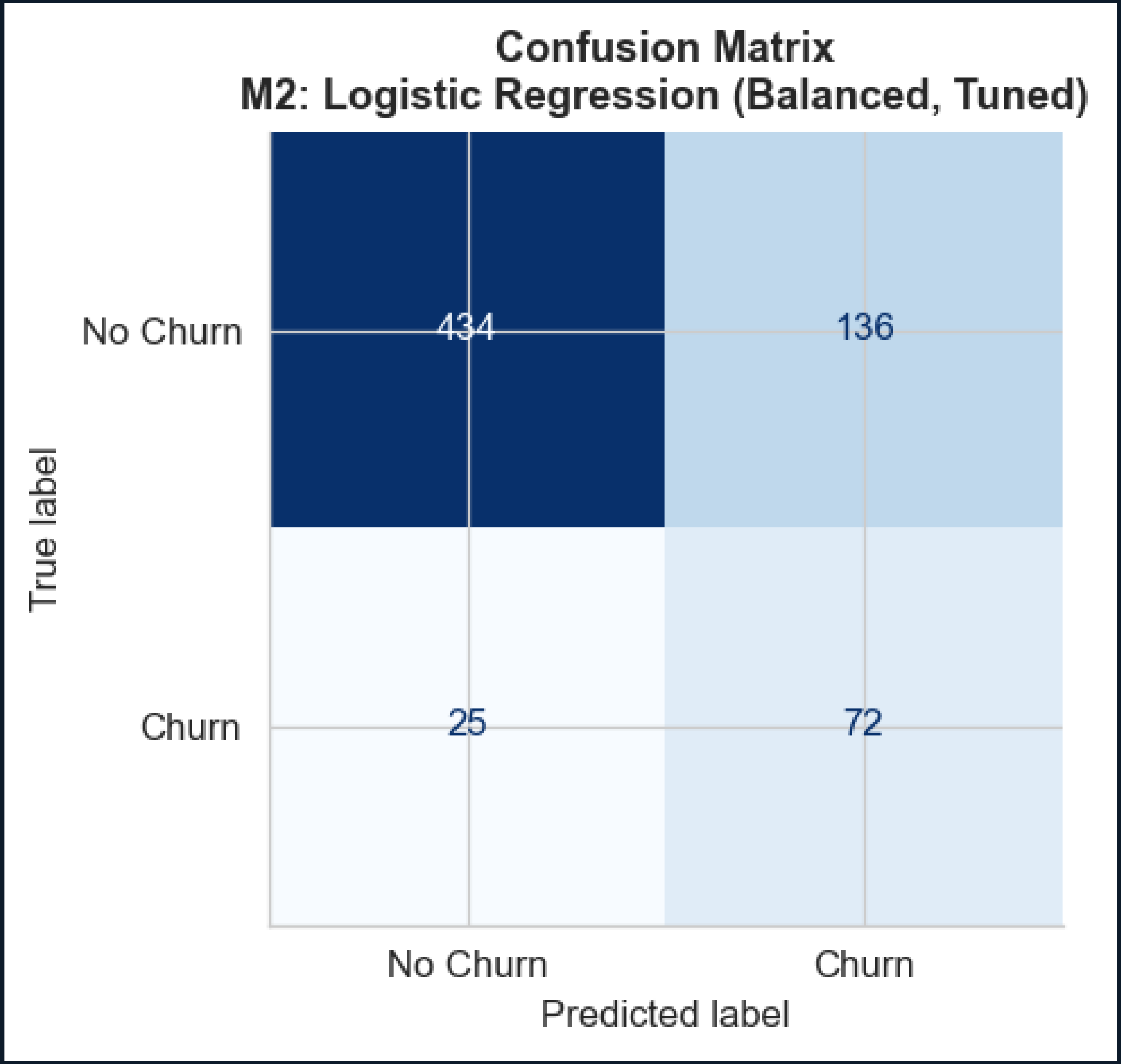
EXPLORATORY DATA ANALYSIS



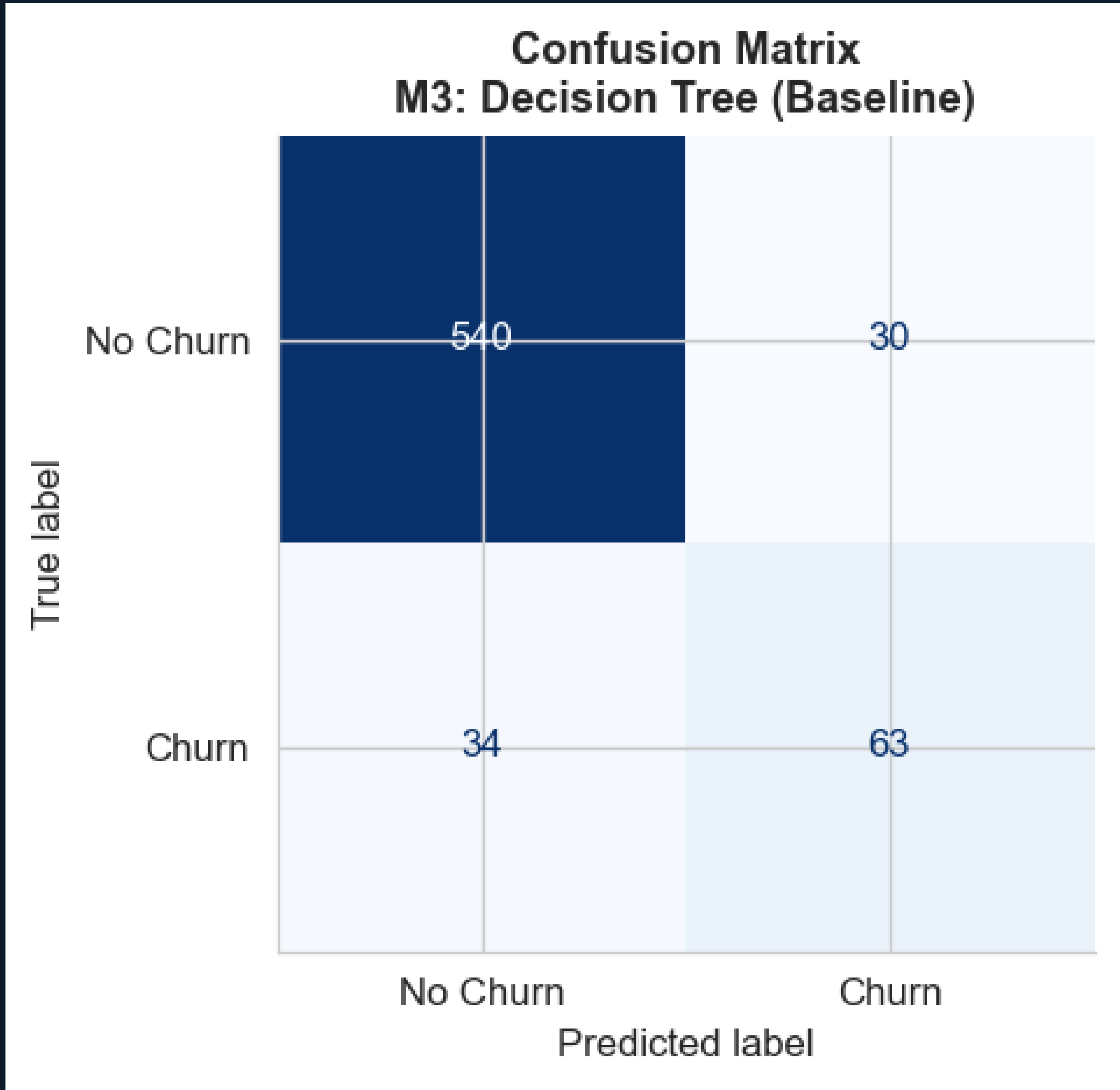
Visual 6 — Correlation Heatmap Charge columns are perfectly collinear (r=1.00) with their minute equivalents — all four charge columns were dropped.



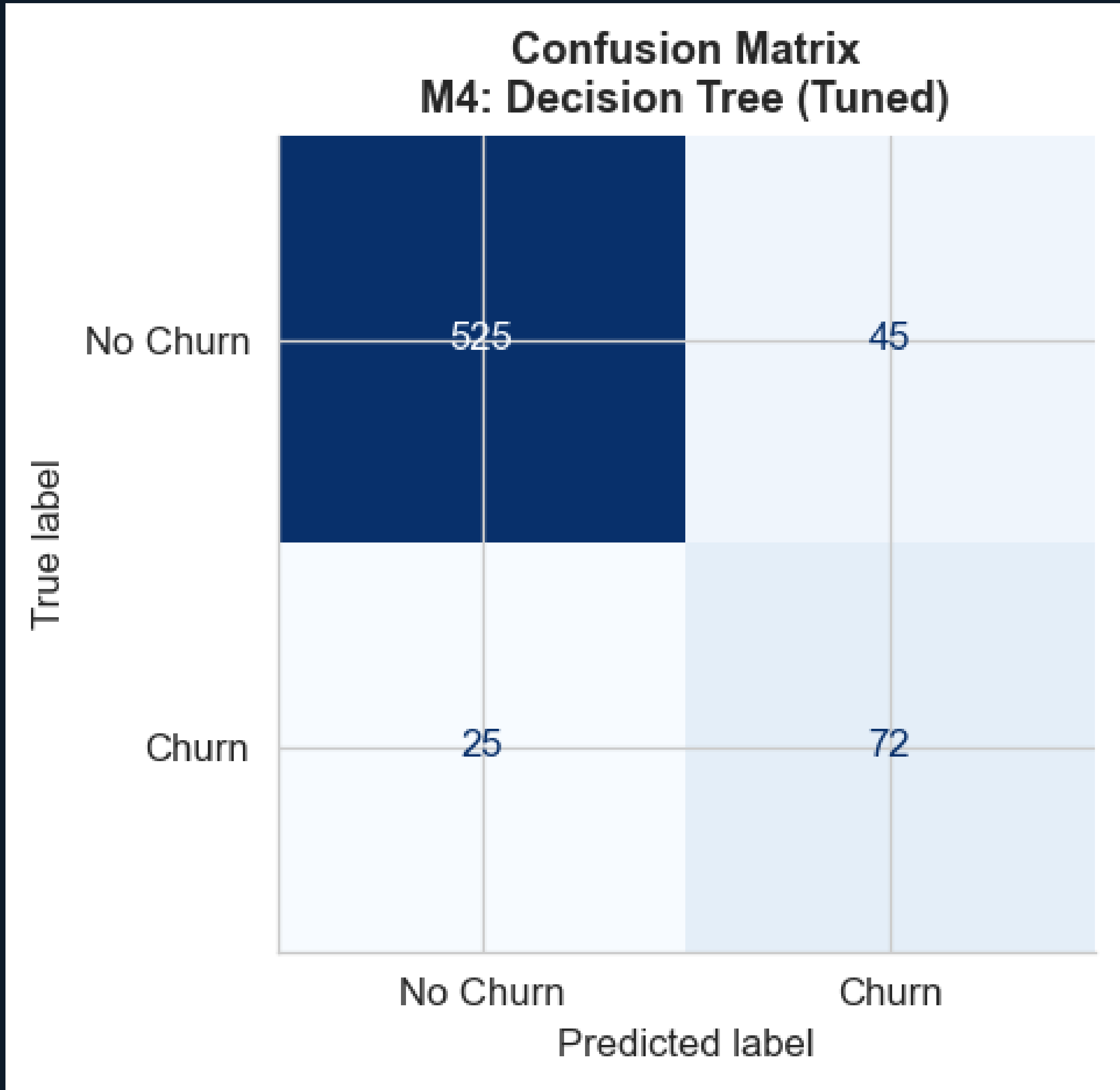
Visual 7 — M1: Logistic Regression (Baseline) — Confusion Matrix Only 24.7% recall. Default threshold badly calibrated for the imbalanced class.



Visual 8 — M2: Logistic Regression (Tuned) — Confusion Matrix class_weight='balanced' + GridSearchCV lifts recall to 74.2%, a 50-point improvement over M1.

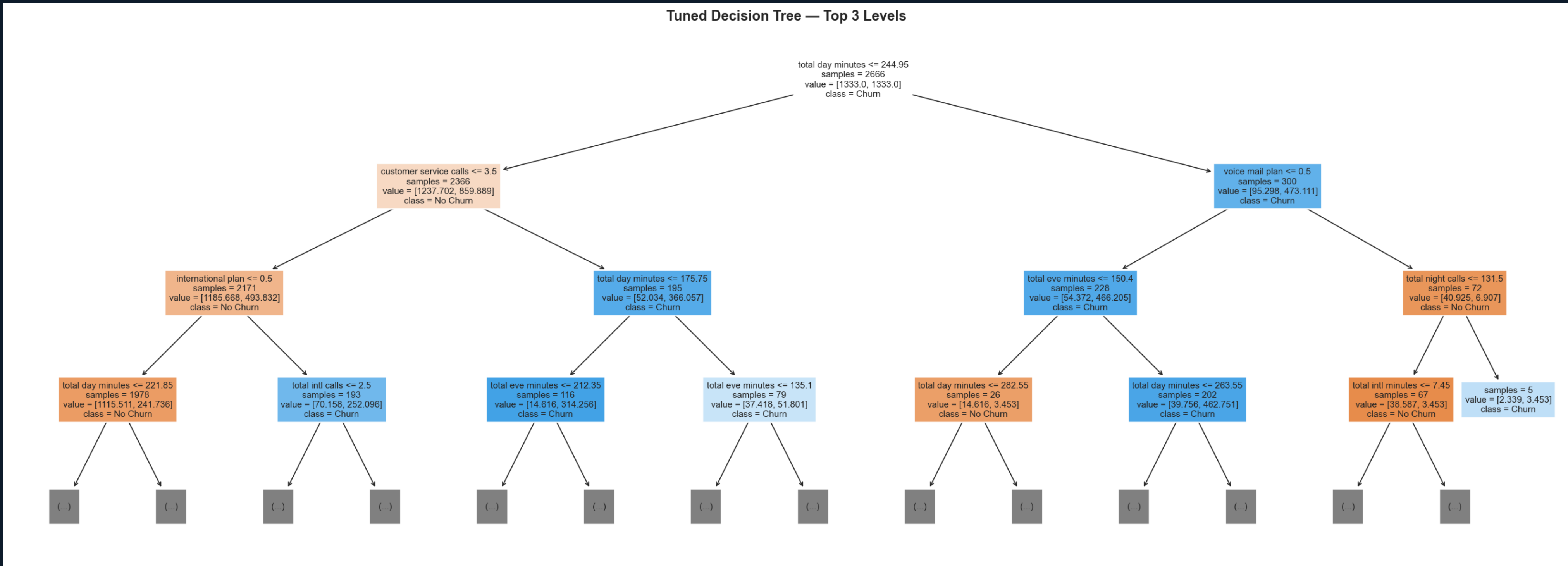


Visual 9 — M3: Decision Tree (Baseline) — Confusion Matrix Train recall = 1.00 vs test recall = 0.649 — textbook overfitting. Non-linear approach still beats M2 on F1.

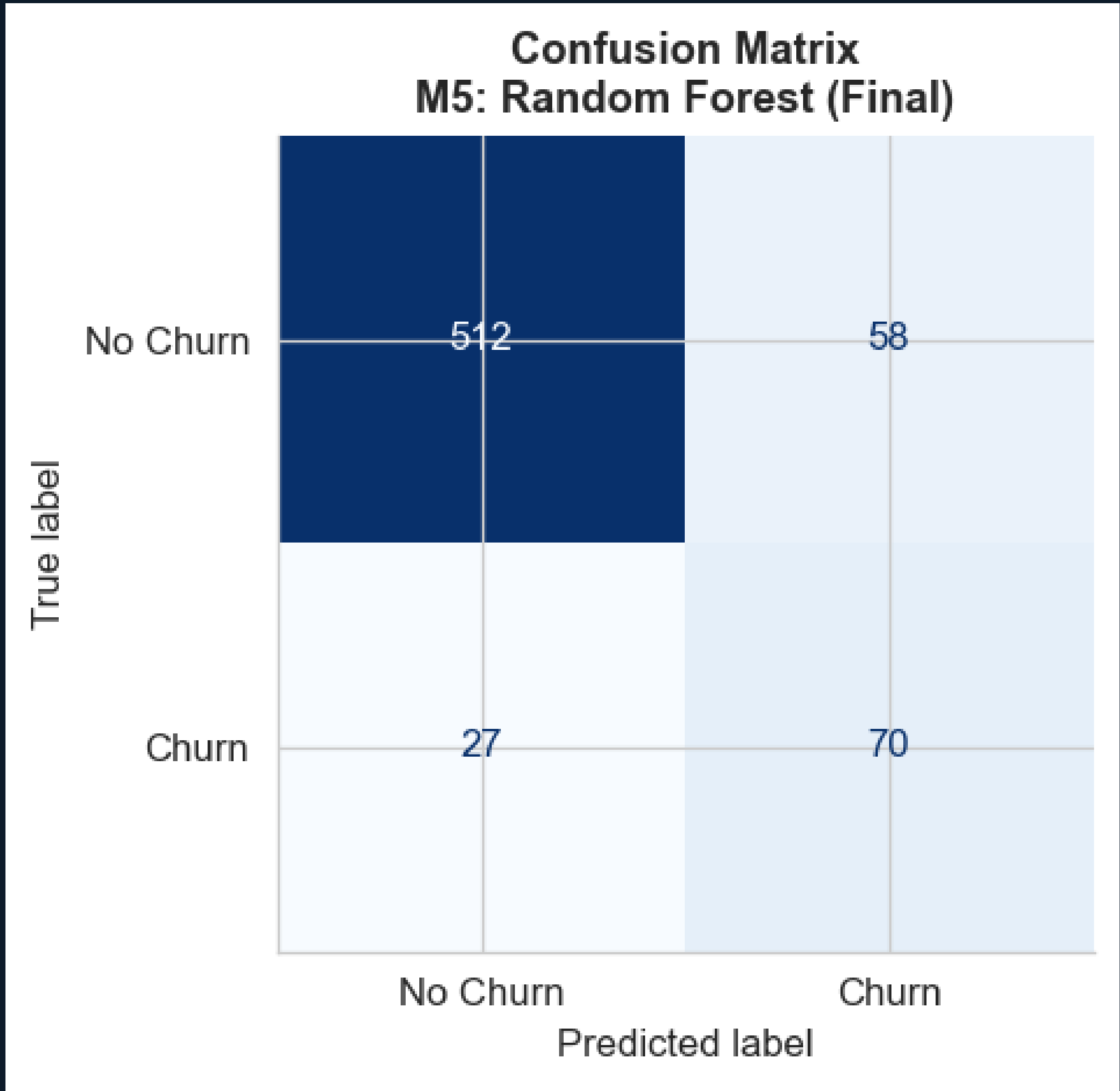


Visual 10 — M4: Decision Tree (Tuned) — Confusion Matrix Overfitting gap shrinks from 0.351 to 0.196. Test recall 74.2%, F1 = 0.673 vs 0.474 for M2.

MODELING

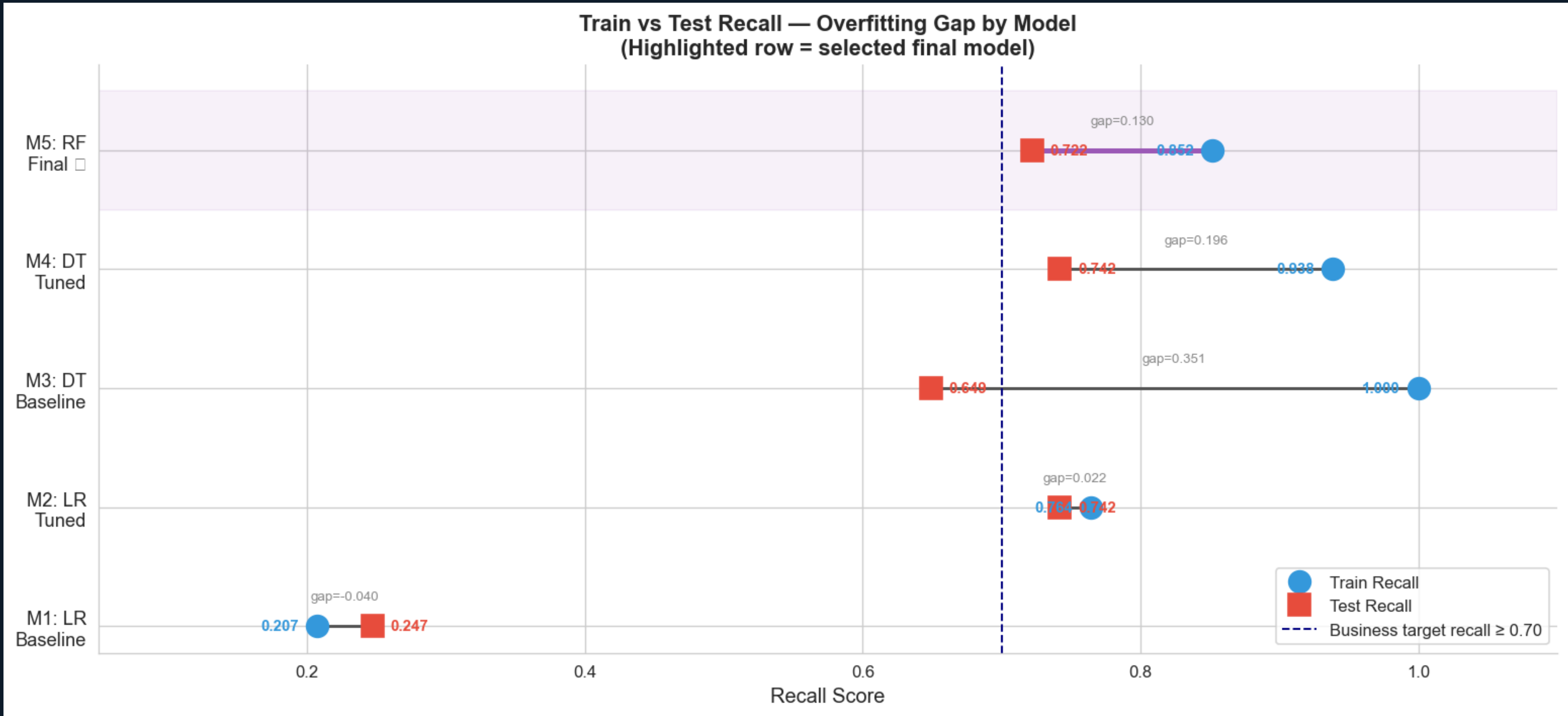


Visual 11 — M4: Decision Tree (Tuned) — Tree Structure First split on customer service calls, confirming EDA insight. Max depth constrained via GridSearchCV.

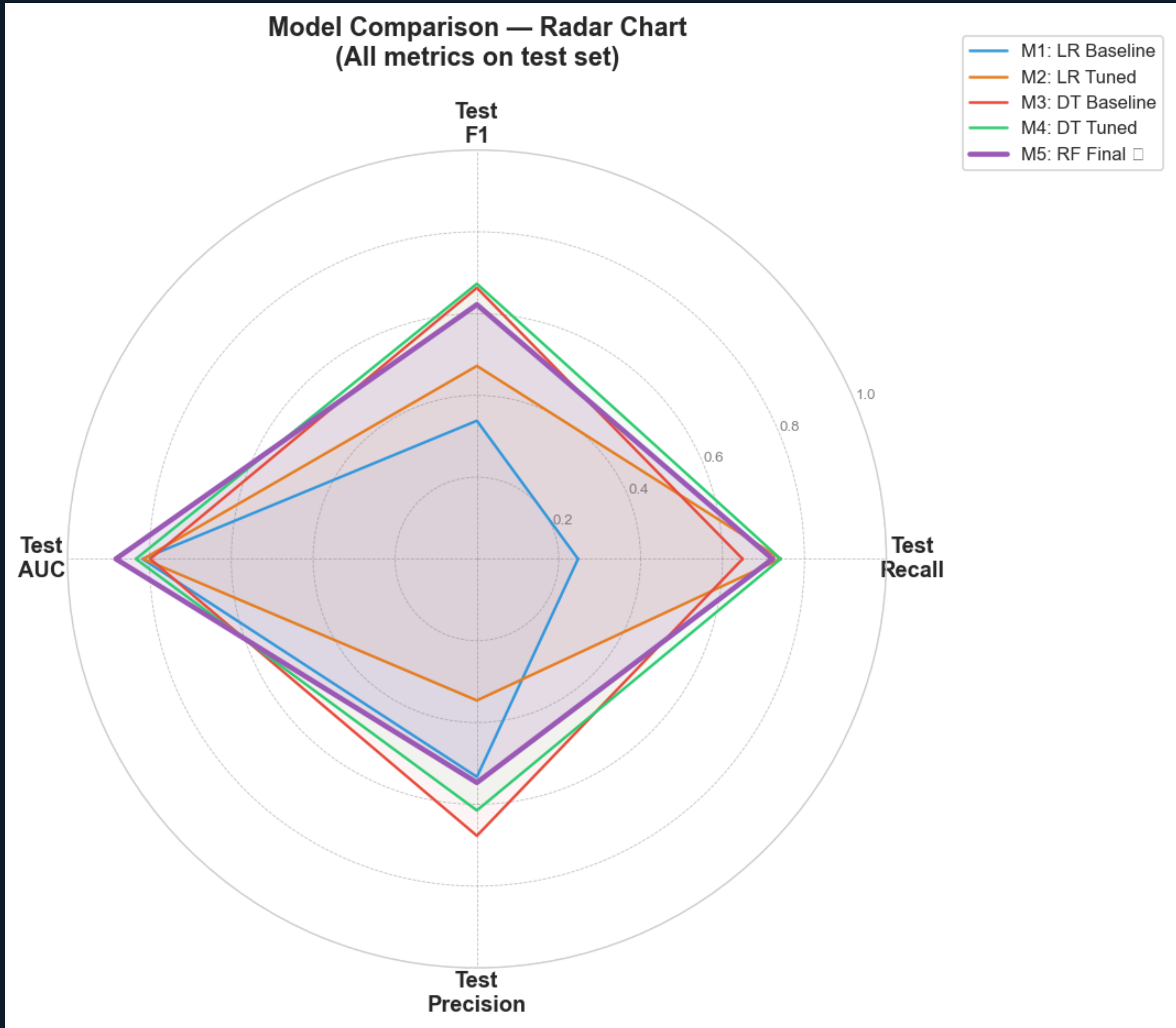


Visual 12 — M5: Random Forest (Final Model) — Confusion Matrix Test recall 72.2%, F1 = 0.622, ROC-AUC = 0.882. Smallest train-test gap (0.130) of all five models.

■■ EVALUATION ■■

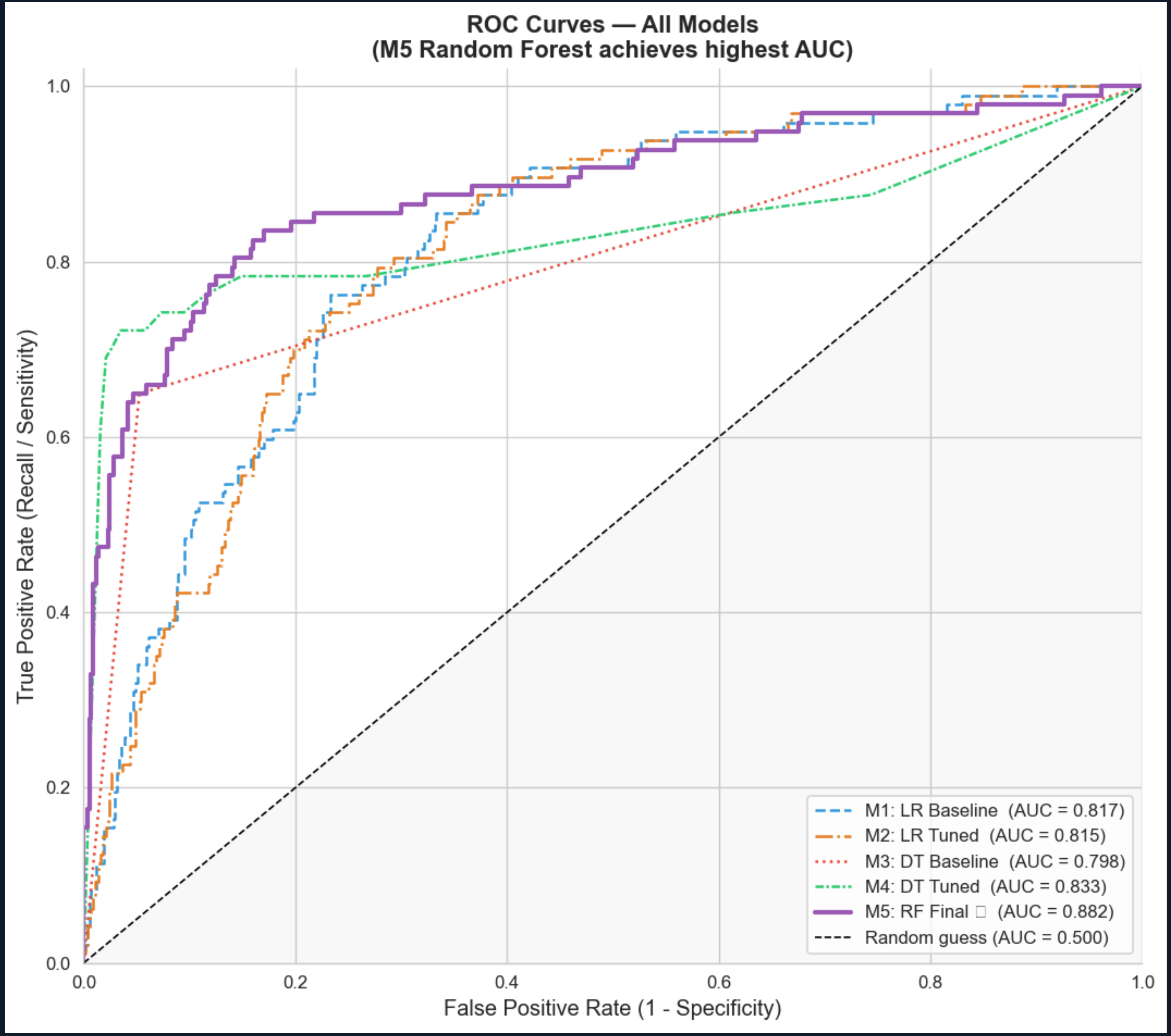


Visual 13 — Dumbbell Chart: Train vs Test Recall Gap Distance between dots shows overfitting. Random Forest has the smallest gap — best generalisation.

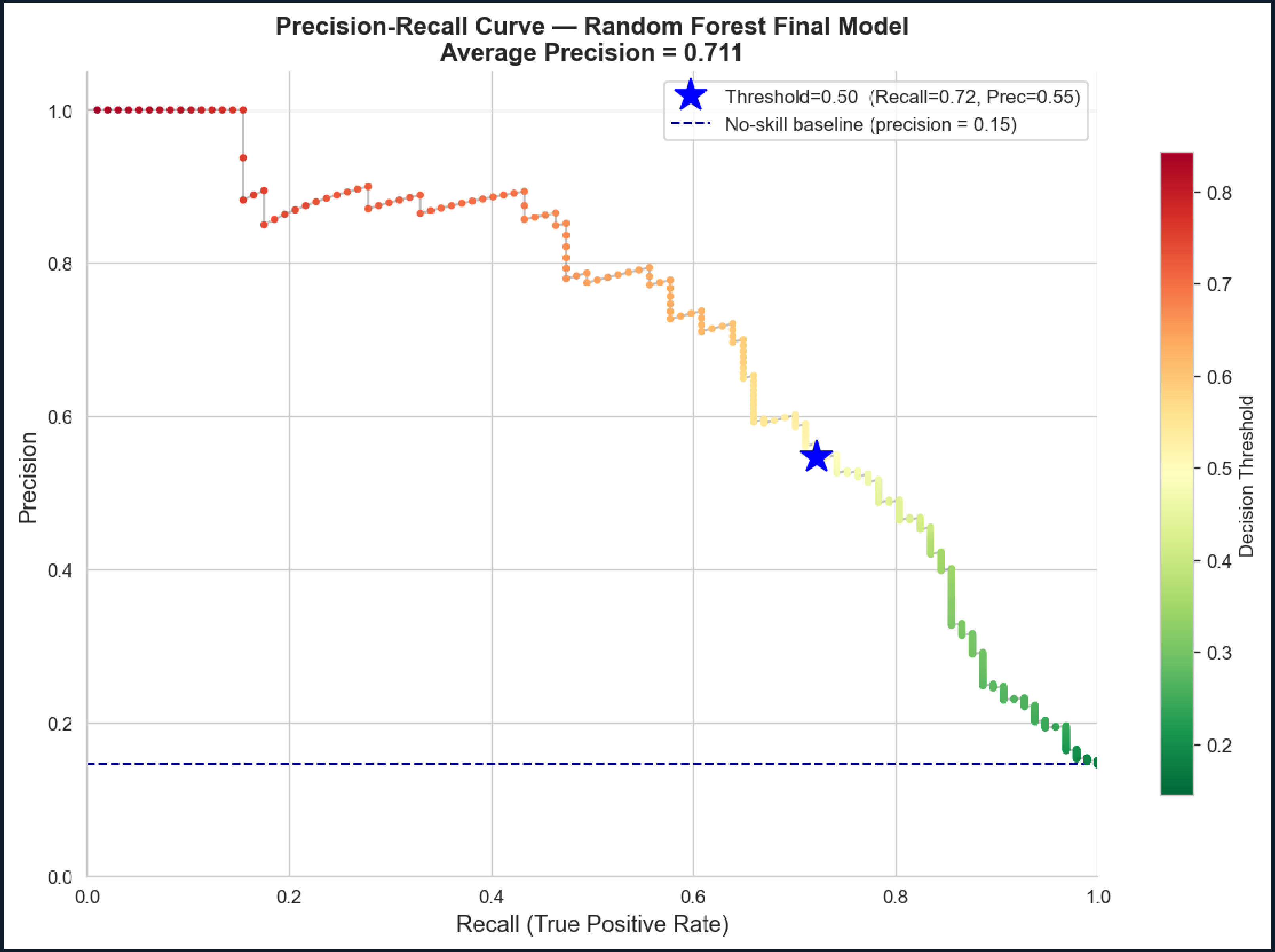


Visual 14 — Radar Chart: All Metrics Across All Models Random Forest (highlighted) is the most well-rounded model across Recall, F1, AUC, and Precision.

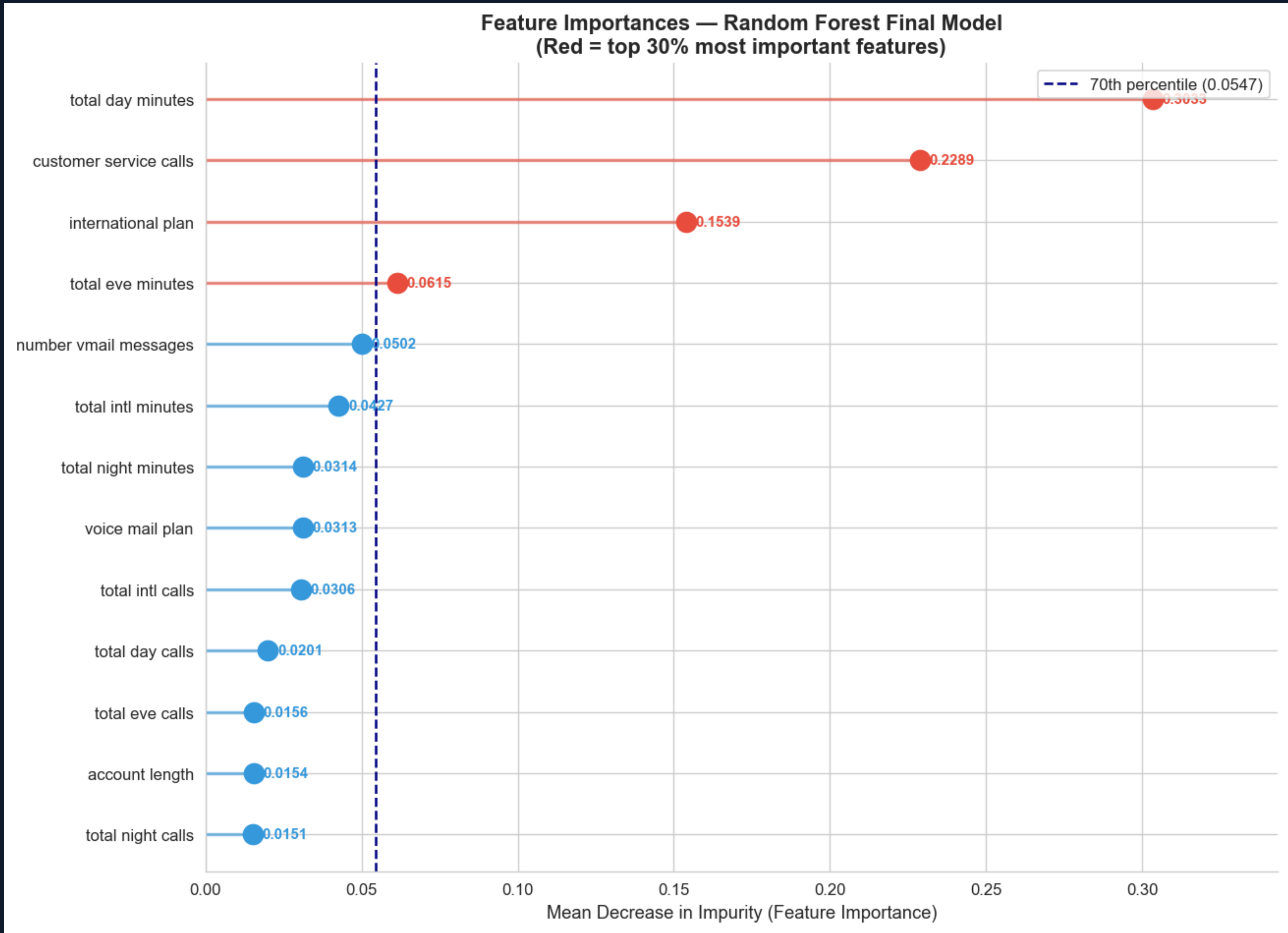
EVALUATION



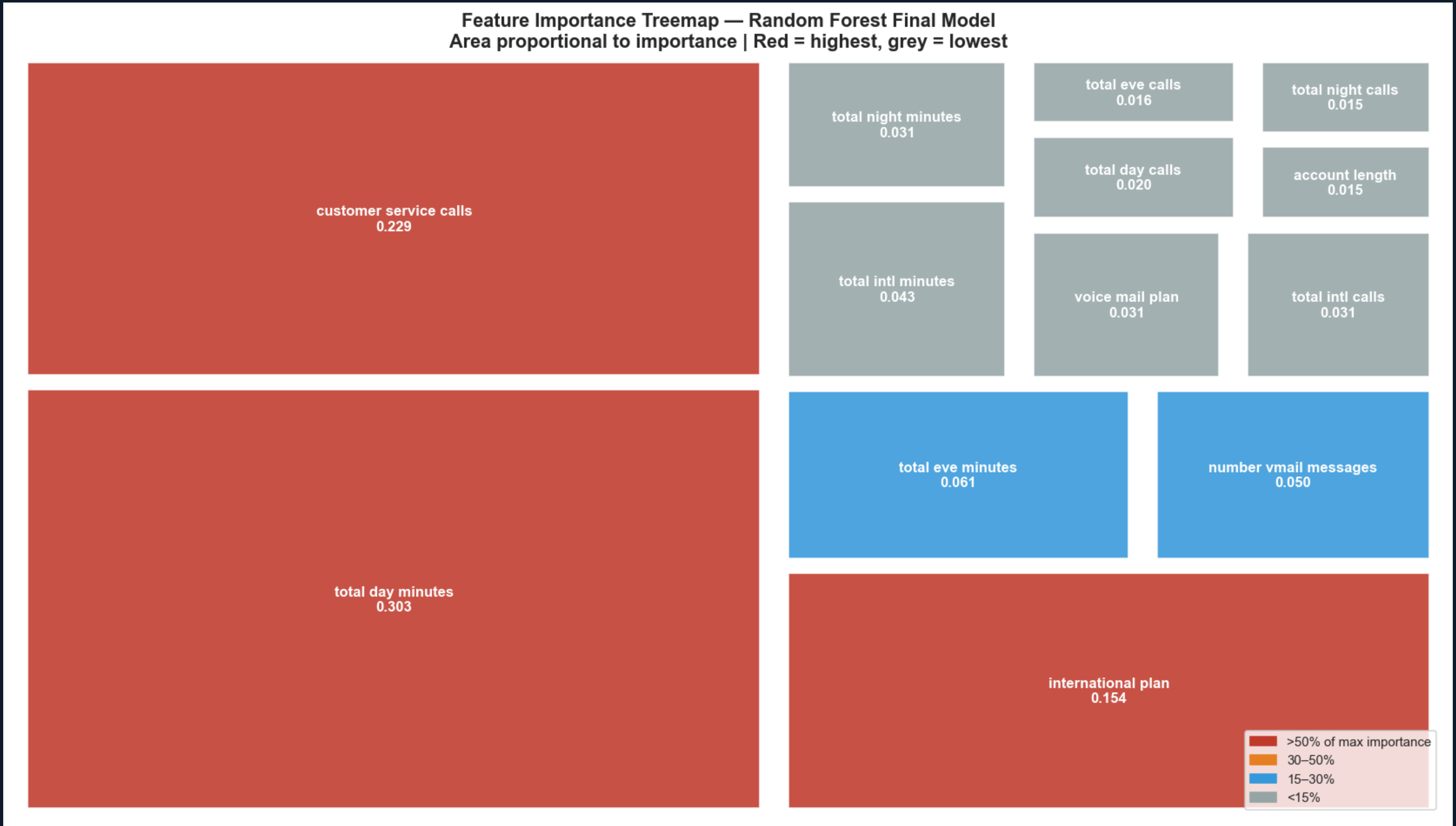
Visual 15 — ROC Curves: All Five Models Random Forest achieves highest AUC (0.882). ROC shows ranking quality at every possible threshold.



Visual 16 — Precision-Recall Curve: Final Model Colour-coded by threshold. Team can move along this curve based on outreach capacity each week.



Visual 17 — Feature Importance: Lollipop Chart Total day minutes (0.303) and customer service calls (0.229) are the top two churn drivers.



Visual 18 — Feature Importance: Treemap Proportional area shows relative feature contribution. Top 3 features account for >68% of importance.